Float and Sink: One Leaf, Both Processes

Lab · Canvas Original #41 · Module 3, Lesson 3 · LT 3.2 (rate of photosynthesis) and LT 3.4 (how photosynthesis and respiration connect)

Name _____ Period _____ Date _____ Role: timer · counter · recorder · materials (circle one)

Question. A leaf in the light gives off oxygen. Does the same leaf also use oxygen, and how would you see it?
An English ivy leaf is full of air spaces. Pull the air out with a syringe and fill the spaces with a little baking-soda water (it supplies carbon dioxide), and a small disk of leaf sinks. In light, its chloroplasts make sugar and release oxygen; the oxygen fills the spaces again and the disk rises. Every living leaf cell also carries out cellular respiration, day and night, using oxygen.

Safety and materials

Goggles on. The syringe has no needle; keep a finger over the tip while you pull. The solution is baking soda, water and dish soap, and it goes down the sink when you finish. Ivy sap can bother some people's skin: wash your hands after the lab, and never eat any part of the plant.

Per group: English ivy leaves, a single-hole punch, a 10 mL or larger plastic syringe, a cup of baking-soda solution (about 1/8 teaspoon in 300 mL water plus 1 drop of soap), a dropper of dish soap, three clear cups labeled A, B and C, a lamp, a ruler, the class dark box, a timer.

Part 1 · Sink the disks, 10 at a time (12 minutes)

1. Punch 35 disks from the leaves, between the big veins. Keep only clean, whole circles.
2. Pull the plunger out, drop 10 disks into the syringe barrel, push the plunger back in until little air is left, and draw up about 5 mL of solution. Tap so the disks float free.
3. Point the tip up, push out the air, then seal the tip with a finger and pull the plunger back hard for 10 to 15 seconds while swirling. Bubbles leave the disks.
4. Release, tap, and repeat until all 10 sink. Ivy is waxy and often needs 4 to 6 pulls; if the disks still float after 3 pulls, add 2 drops of soap to your solution and draw it up again.
5. Pour the 10 sunk disks into cup A with about 3 cm of solution. Sink the next 10 for cup B, then 10 for cup C. Take out any disk that floats.

Part 2 · Light near, light far, no light (15 minutes)

Cup A sits 10 cm from the lamp. Cup B sits 30 cm from the lamp. Cup C goes in the dark box. Start the timer when the lamp goes on, and count the disks touching the surface in A and B every minute. Tap each cup before you count so no disk clings to the wall. Peek into C only at 8 and 15 minutes.
Predict before you start: which cup will have half its disks floating first, and which may never get there?

Minute	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A · 10 cm																
B · 30 cm																
C · dark																

Number of disks floating (out of 10). Grey boxes: leave cup C closed. Enter your numbers in your class's Canvas lab data form as you go if your teacher uses one.

Part 3 · Lights out for cup A (rest of class)

When at least 8 disks in cup A float, move cup A into the dark box. Count its floating disks every 2 minutes for 10 minutes, then once more just before the end of class. Ivy can be slow to sink, so move cup A as soon as it reaches 8.
Predict: what will the floating disks in cup A do in the dark, and why? _____

Minutes in the dark	0	2	4	6	8	10	End of class
Cup A disks floating							

Part 4 · Pool the class data

For each cup, record the minute when 5 of the 10 disks were floating (the half-float time). Write “never” if it did not happen in 15 minutes. Copy your numbers to the board, then copy the class average. Your teacher runs one more cup at 10 cm: disks sunk in plain water and soap, with no baking soda.

	Our half-float time (min)	Class average (min)	Groups that never reached half
A · 10 cm			
B · 30 cm			
C · dark			
Water only (teacher's cup)			

Graph the floating disks against time for cups A and B on the grid your teacher provides, or on GoodNotes: time on the x-axis, disks floating on the y-axis, a key for each line.

Part 5 · Make sense of it

1. What made the disks in cup A rise? Name the gas, the process and the organelle.

2. Compare cups A and B. What does the difference tell you about light and the rate of photosynthesis? Use your half-float times.

3. Cup C sat in the same solution with the same carbon dioxide. Why did its disks stay down?

4. The teacher's water-only cup sat in the light with no baking soda. What did its disks do, and what does that tell you about what photosynthesis needs?

5. In Part 3, what happened to cup A's disks in the dark? Name the process that used the oxygen, and the organelle where it happens.

6. Complete both lines. In the light, the leaf disk makes oxygen (faster / slower) than it uses it. In the dark, the disk (makes / uses) oxygen only.

7. Matter cycles and energy flows. Name one atom that cycled through this lab, and trace where the energy went from the lamp onward.

Advanced and P3: design your own run. Change one variable (distance, baking soda amount, a green or red filter over the lamp, or a sun leaf against a shade leaf), keep everything else the same, and predict before you test. Variable changed: _____ Controlled: _____

Take it further: read **Dawn in Cobbs Creek** (Biology #20, scienceonepagers.org), where the same day-and-night pattern plays out in a whole creek. Absent on lab day? Run the leaf-disk simulation in Biochemistrypedia, “Capturing Sunlight” (Bio Tool #30).

Canvas Original #41 · An original lab sheet built on the classic floating leaf-disk method, run with English ivy (revised 27 October 2026). © J. R. Schwebach, CC BY-NC 4.0 · scienceonepagers.org